

CSE 321– Operating Systems
QUIZ 2
Summer Semester – 2022

Grade

Full Name (in Block Letter): _____

ID: _____ **Section:** _____ **Signature:** _____

Marks: 20 points

Time: 35 minutes

Instructions: Answer all questions on the space provided below for each.

Question 1: CO3 (1+ [4+2+2] points): a. What do you understand by the term “convoy effect”?
b. Consider the following set of processes with the length of the CPU-burst time given in milliseconds.
Draw the Gantt Charts illustrating the execution of these processes using preemptive priority (a biggest number implies a higher priority). **Find** average waiting time average turnaround time for below data set.

Process	Arrival Time	Priority	Burst Time
P1	0	2	4
P2	1	4	2
P3	2	6	3
P4	3	10	5
P5	4	8	1
P6	5	12	4
P7	6	9	6

Answer:

Question 2 CO3 (2+2 points): a. Write one possible advantages and disadvantages of RR scheduling algorithm.

b. “Multilevel feedback queue prevents starvation”? Justify your answer.

Answer:

Question 3 CO3 (4+2+1 points): Draw a Gantt chart and illustrating the execution of the process using Round Robin scheduling algorithm (time quantum = 8 s,) .Calculate average waiting time, and number of context switching.

Processes	Arrival Time	Priority	Burst Time
P1	18	1	40
P2	29	1	17
P3	0	1	28
P4	21	1	37
P5	12	1	31

Answer: